

Pengkun Jiao

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1 EDUCATION

East China Normal University

Sep 2017 – Jun 2021

B.Eng. in Computer Science and Technology

Fudan University

Sep 2021 – Jun 2026 (Expected)

Ph.D. in Artificial Intelligence

Research Interests: Multimodal Learning, Large Language Models, Reinforcement Learning, Efficient Model Architecture

Honors: Venustech Corporate-Sponsored Scholarship

2 RESEARCH EXPERIENCE

Alibaba Group – Taobao & Tmall Group

May 2025 – Sep 2025

Research Intern

- Investigated post-training reinforcement learning for large models in relevance reasoning scenarios, focusing on step-level reward modeling, step-level credit assignment, and hint-based RL to address sparse reward issues.

Singapore University of Technology and Design (SUTD)

Apr 2023 – Nov 2023

Research Assistant

- Conducted research on enhancing open-vocabulary 3D object detection using vision-language foundation models.

3 RECENT RESEARCH PROJECTS

- Head-Level KV Cache Eviction via Long-Term Marginal Utility Optimization:** Proposed LU-KV, the first KV cache compression framework targeting long-term marginal utility of eviction decisions. Achieves near-optimal head-level budget allocation via convex-hull relaxation and a marginal-utility-driven greedy solver, compressing KV cache by 80% with minimal performance degradation (**ICML 2026**).
- Reinforcement Learning for Post-Training LLMs in Correlation Reasoning:** Investigated step-level reward models, step-level credit assignment, and hint-based RL to address sparse reward issues and inefficient learning on hard samples in GRPO-style RL strategies (**WWW 2025**).
- Vision-Related Attention Reallocation for Mitigating Language Misleading:** Analyzed multimodal attention allocation anomalies (e.g., Attention Sink) and proposed a training-free visual attention reallocation strategy to suppress misleading token interference, reducing hallucinations in multimodal large language models (**Under Review**).
- Efficient Multimodal LLM Fine-Tuning via Dual Low-Rank Subspace Adaptation:** Proposed Dual-LoRA, a dual low-rank adaptation paradigm combined with dynamic convolution for enhanced local visual perception, surpassing LoRA-MoE systems with $1.59\times$ runtime using only $1.14\times$ standard LoRA runtime across multiple benchmarks (**ICCV 2025**).
- Dynamic Activation with Heterogeneous Mixture-of-Experts for Multi-Task Fine-Tuning:** Introduced linear rectified routing for dynamic sparse expert activation and heterogeneous expert group configurations to address data conflicts and task-complexity mismatch in fixed-routing LoRA-MoE. Achieved state-of-the-art results on multiple downstream tasks under equivalent parameter budgets (**ICMR 2026**).

4 SELECTED PUBLICATIONS

- P. Jiao**, B. Zhu, J. Chen, C.-W. Ngo, Y.-G. Jiang.
“From Holistic to Localized: Local Enhanced Adapters for Efficient Visual Instruction Fine-Tuning.”
IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
- P. Jiao**, N. Zhao, J. Chen, Y.-G. Jiang.
“Unlocking Textual and Visual Wisdom: Open-Vocabulary 3D Object Detection Enhanced by Comprehensive Guidance from Text and Image.”
European Conference on Computer Vision (ECCV), 2024.
- P. Jiao**, N. Zhao, J. Chen, Y.-G. Jiang.
“Domain Expansion and Boundary Growth for Open-Set Single-Source Domain Generalization.”
IEEE Transactions on Multimedia (TMM), 2024.
- P. Jiao**, X. Wu, B. Zhu, J. Chen, C.-W. Ngo, Y.-G. Jiang.
“Rode: Linear Rectified Mixture of Diverse Experts for Food Large Multi-modal Models.”
ACM International Conference on Multimedia Retrieval (ICMR), 2026.
- Z. Tang*, **P. Jiao***, X. Chen, W. Liu, S. Li, J. Chen.
“Predicting Future Utility: Global Combinatorial Optimization for Task-Agnostic KV Cache Eviction.”
International Conference on Machine Learning (ICML), 2026.

6. Z. Tang*, **P. Jiao***, B. Zhu, H. Qi, J. Chen, Y.-G. Jiang.
“Spatiotemporal Sycophancy: Negation-Based Gaslighting in Video Large Language Models.”
Annual Meeting of the Association for Computational Linguistics (ACL) Findings, 2026.
7. **P. Jiao***, B. Zhu, J. Chen, C.-W. Ngo, Y.-G. Jiang.
“Don’t Deceive Me: Mitigating Gaslighting through Attention Reallocation in LMMs.”
Under Review.